
ME 607: Materials Science For Failure Analysis

Course Code	ME 607
Course Name	Materials Science For Failure Analysis
L-T-P-C	3-0-0-3
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- Introduction remarks on Materials Science in the context of Engg.:
- Structure of perfect and imperfect solids; Elastic deformation and stress distribution, stress-strain relations under uniaxial loading.
- Introduction, theoretical strength of crystals and the motion of dislocation, energy of a dislocation and stable Burgers vectors.
- Slip planes and slip systems, relation between dislocation movement and plastic flow, dislocation generation, other modes of Deformation in crystalline solids.
- Some strengthening mechanisms.
- The phenomenon of yield point and strain hardening. Theories of yielding and strain hardening.
- Recovery, mechanisms of deformation at elevated temperatures, creep.
- Mechanism of fracture. Ductile - Brittle transition, fracture Design criteria for materials, environmental effects.
- Mechanical behavior of engineering materials under fatigue.
- Selection of materials and Processes, case studies.

Text and Reference Books

1. George E. Dieter, **Mechanical Metallurgy**, McGraw Hill Book Company.
2. R. W. K. Honeycombe, **Plastic Deformation of Metals**, EWP
3. William D. Callister Jr., **Materials Science and Engineering**, Wiley India (P) Ltd.
4. Knott, **Fundamentals of Fracture Mechanics**.
5. A. H. Cottrell, **Mechanical Properties of Matter**, Wiley
6. K. J. Brown, **Introduction to Mechanical Behaviour of Materials**, Wiley.